

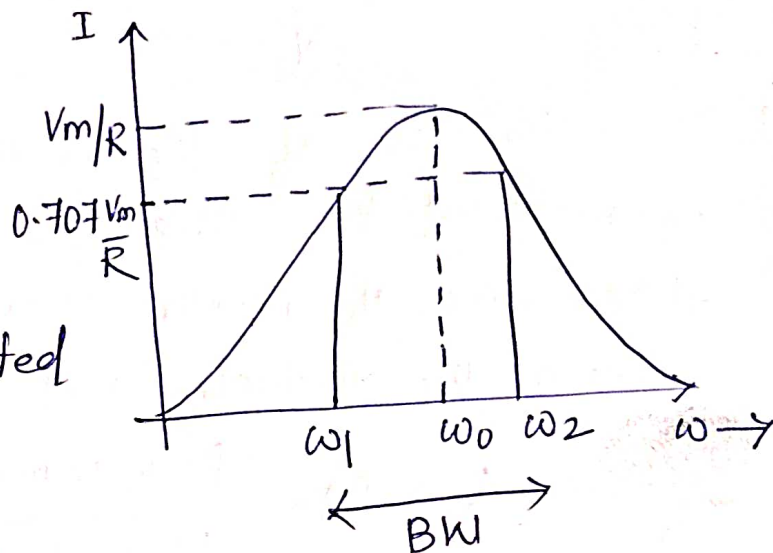
Bandwidth - of SERIES RLC CKT

Module 2

$$|I| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

$$P(\omega) = \frac{1}{2} I^2 R$$

The highest - power dissipated occurs at Resonance when $I = V_m/R$



$$P(\omega_0) = \frac{1}{2} \frac{V_m^2}{R}$$

At certain frequency $\omega = \omega_1, \omega_2$, the dissipated power is half the maximum value -

$$P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R}$$

ω_1 and ω_2 are called as the half power frequencies

The half power frequencies are obtained by setting Z equal to $\sqrt{2}R$ ✓

$$\sqrt{R^2 + (\omega L - 1/\omega C)^2} = \sqrt{2}R$$

$$\omega_1 = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\boxed{\omega_0 = \sqrt{\omega_1 \omega_2}}$$

$$B = \omega_2 - \omega_1$$

(*) The "sharpness" of the resonance in a resonant ckt is measured by Quality factor Q .
At resonance, the reactive energy in the circuit oscillates between the inductor and the capacitor.

$$Q = 2\pi \cdot \frac{\text{Peak or max energy stored}}{\text{Energy dissipated per cycle or Energy dissipated in one period at resonance.}}$$

$$Q = 2\pi \cdot \frac{\frac{1}{2} LI^2}{\frac{1}{2} I^2 R (1/f_0)} = \frac{2\pi f_0 L}{R}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

$$\omega_0^2 RC = B$$

$$\omega_1 = \omega_0 - \frac{B}{2}$$

$$\omega_2 = \omega_0 + \frac{B}{2}$$



bandwidth : (BW) : $\rightarrow$

for a series R-L-C circuit,

bandwidth is defined as, $I_0 = \frac{V}{R}$

the range of frequencies

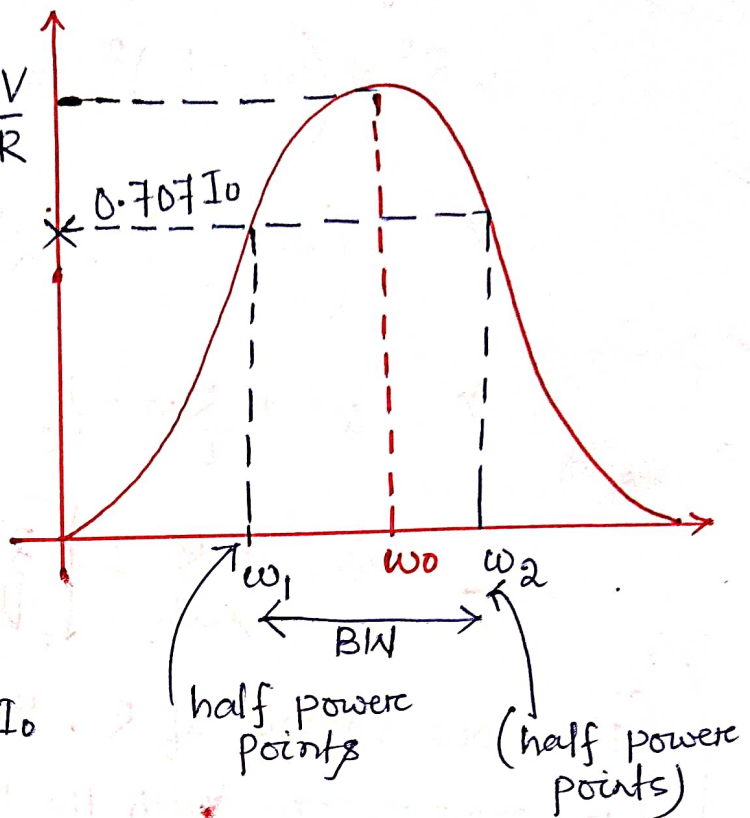
for which the power delivered to R $\geq \frac{P_0}{2}$, where P_0 is

the power delivered to R at resonance.

$$I_1^2 R = I_2^2 R = \frac{1}{2} I_0^2 R$$

$$I_1 = I_2 = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

$$BW = \omega_2 - \omega_1$$



EXPRESSION FOR BANDWIDTH : $\rightarrow$

Generally at any frequency ' ω '

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \rightarrow (1)$$

At half power points -

$$I = \frac{I_0}{\sqrt{2}}$$

$$\text{But } I_0 = \frac{V}{R} \quad \text{so, } I = \frac{V}{\sqrt{2} R} \rightarrow (2)$$

For Equations (1) and (2)

$$\frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} = \frac{V}{\sqrt{2} R}$$

$$\text{so, } \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2} R$$

Squaring both the sides,

$$\Rightarrow R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

$$\Rightarrow \left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$$

$$\Rightarrow \omega^2 \pm \frac{R}{L}\omega - \frac{1}{LC} = 0$$

$$\omega = \pm \frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

For low values of R , the term $\left(\frac{R^2}{4L^2}\right)$ can be neglected in comparison with the term $\frac{1}{LC}$.

$$\omega = \pm \frac{R}{2L} \pm \sqrt{1/LC}$$

$$= \pm \frac{R}{2L} \pm \frac{1}{\sqrt{LC}}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\Rightarrow \omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

We know that $f_0 = \frac{1}{2\pi\sqrt{LC}}$; $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\boxed{\begin{aligned}\omega_0 &= \sqrt{\omega_1 \omega_2} \\ BW &= \omega_2 - \omega_1\end{aligned}}$$

(*) The "sharpness" of the resonance in a resonant ckt is measured by Quality factor

$$Q = 2\pi \cdot \frac{\text{Peak or max energy stored}}{\text{Energy dissipated per cycle or Energy dissipated in one period at resonance}}$$

$$Q = \frac{2\pi \cdot \frac{1}{2} LI^2}{\frac{1}{2} I^2 R \left(\frac{1}{f_0} \right)} = \frac{2\pi f_0 L}{R}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$$

$$BW = \omega_2 - \omega_1 = \frac{R}{L}$$

$$BW = f_2 - f_1 = \frac{R}{2\pi L}$$

$$\begin{aligned} \omega_1 &= \omega_0 - \frac{BW}{2} = \omega_0 - \frac{R}{2L} \\ \omega_2 &= \omega_0 + \frac{BW}{2} = \omega_0 + \frac{R}{2L} \end{aligned}$$

$$\begin{cases} f_1 = f_0 - \frac{R}{4\pi L} \\ f_2 = f_0 + \frac{R}{4\pi L} \end{cases}$$

OR

$$QF = \frac{\text{Voltage across inductor or capacitor}}{\text{Voltage at Resonance}} = \frac{V_{L0}}{V} = \frac{V_{C0}}{V}$$

$$Q = \frac{I_0 X_{L0}}{I_0 R} = \frac{X_{L0}}{R} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{\left(\frac{1}{\sqrt{LC}} \right) L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

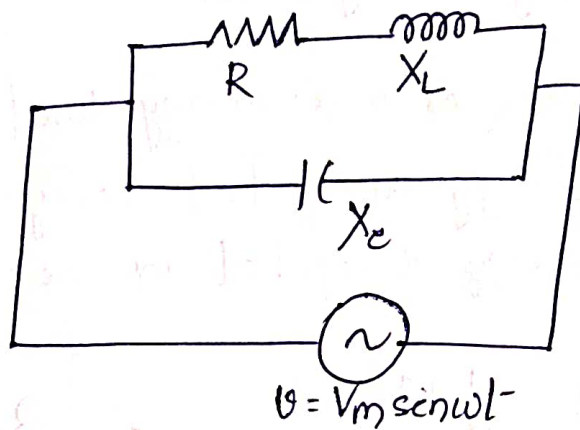
Parallel Resonance $\Rightarrow$

$$\bar{Z}_1 = R + jX_L$$

$$\bar{Z}_2 = -jX_C$$

$$\bar{Y}_1 = \frac{1}{\bar{Z}_1} = \frac{1}{R + jX_L}$$

$$= \frac{R - jX_L}{R^2 + X_L^2}$$



$$\bar{Y}_2 = \frac{1}{\bar{Z}_2} = \frac{1}{-jX_C} = \frac{j}{X_C}$$

Total Admittance of the circuit

$$\bar{Y} = \bar{Y}_1 + \bar{Y}_2$$

$$= \frac{R - jX_L}{R^2 + X_L^2} + \frac{j}{X_C}$$

$$= \frac{R}{R^2 + X_L^2} - j \left(\frac{X_L}{R^2 + X_L^2} - \frac{1}{X_C} \right)$$

At resonance, the ckt is purely resistive

$$\Rightarrow \frac{X_L}{R^2 + X_L^2} - \frac{1}{X_C} = 0$$

$$\Rightarrow \frac{X_L}{R^2 + X_L^2} = \frac{1}{X_C}$$

$$\Rightarrow X_L X_C = R^2 + X_L^2$$

$$\Rightarrow \omega_0 L \frac{1}{\omega_0 C} = R^2 + \omega_0^2 L^2$$

$$\Rightarrow \omega_0^2 L^2 = \frac{L}{C} - R^2$$

$$\Rightarrow \omega_0^2 = \frac{1}{L^2} \left(\frac{L}{C} - R^2 \right)$$

$$= \frac{1}{LC} - \frac{R^2}{L^2}$$

$$\boxed{\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}}$$

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

→ If R is very small compared to L

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

⇒ Behaviour of conductance G , Inductive susceptance B_L and capacitive susceptance with change in freq:

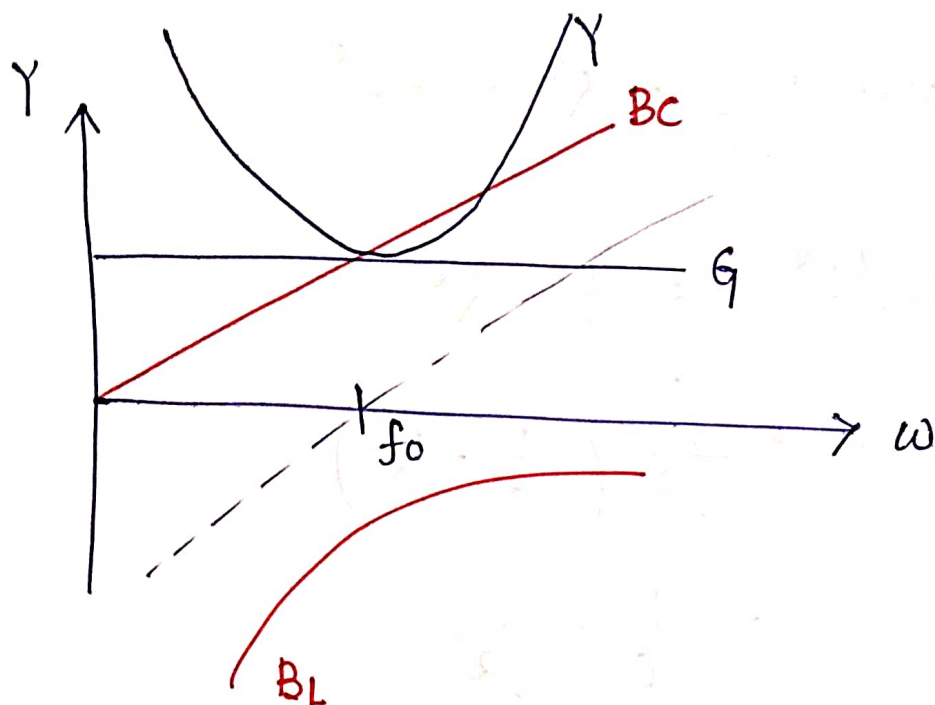
Conductance remains constant with change in frequency

$$\text{Inductive susceptance } B_L = \frac{1}{jX_L} = \frac{-j}{X_L} = \underline{\underline{\frac{-j}{2\pi L}}}$$

$$\text{Capacitive susceptance } B_C = \frac{1}{-jX_C} = \frac{j}{1/X_C} = j2\pi f C$$

$$X_C = \frac{1}{j\omega C}$$

$$\underline{\underline{B_C = j\omega C = j2\pi f C}}$$



(a) $f < f_0$ inductive susceptance dominates -
Hence current lags behind voltage
Pf = Lagging

(b) $f = f_0$; net susceptance is zero
Hence admittance is minimum
and impedance is maxm. (unity Pf)

(c) $f > f_0$ capacitive susceptance predominates
current leads the voltage
Pf = Leading

QF of parallel Resonant ckt : It is a measure of current magnification in parallel

$$Q = \frac{\text{current through inductor or capacitor}}{\text{current at resonance}}$$

$$= \frac{I_{C0}}{I_0}$$

$$Q = \frac{V/X_{C0}}{V/R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$